

SHIKANG PENG

Department of Psychology, University of Toronto
Rotman Research Institute at Baycrest

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Email: speng@research.baycrest.org
Website: <https://shikangpeng.github.io/>

EDUCATION

- 2024- Ph.D., Psychology – Computational Cognitive Neuroscience
 University of Toronto
 Advisors: Drs. Brian Levine & Brad Buchsbaum
- 2023-24 M.A., Social Sciences – Psychology
 University of Chicago
 Advisor: Dr. Wilma Bainbridge
 Thesis: Predictability of memorability neural pattern and attention network in subsequent memory retrieval
- 2019-23 B.Sc., Psychology with Honors | First Class Honors
 University of Alberta
 Advisor: Dr. Peter Dixon
 Thesis: Text-based and memory-based metrics of cognitive coupling

IN PREP & PREPRINTS

- Peng, S.**, Utochkin, I. S., Levine, B., & Bainbridge, W. A. (In prep). Behavioral change need not imply reorganization of memory structure.
- Peng, S.**, Levine, B., & Buchsbaum, B. R. (In prep). Memory transforms differentiated perceptual features through representational convergence.
- Peng, S.**, Buchsbaum, B. R., & Levine, B. (In prep). Intrinsic hippocampal long-axis connectivity maps onto trait variation in autobiographical memory.
- St Jacques, P., Feminella, J., & **Peng, S.** (Under review). Seeing events in 3D enhances scene memory and vividness. *PsyArXiv*. https://doi.org/10.31234/osf.io/z9thr_v1

PUBLICATIONS

2. **Peng, S.**, & Bainbridge, W. A. (2026). Image memorability predicts social media virality and externally-associated commenting. *Computers in Human Behavior*, 174, 108799. <https://doi.org/10.1016/j.chb.2025.108799>
1. **Peng, S.**, & Dixon, P. (2024). Text-based and memory-based metrics of cognitive coupling. *Canadian Journal of Experimental Psychology*, 78(4), 222-229. <https://doi.org/10.1037/cep0000349>

CONFERENCE PRESENTATIONS

Oral Presentations

1. **Peng, S.**, Levine, B., & Buchsbaum, B. R. (2026, May). Beyond reinstatement: Transformation of cortical feature representations during memory recall. *Toronto Area Memory Group Conference (TAMeG)*, Toronto, ON, CA.

Poster Presentations

7. **Peng, S.**, Utochkin, I. S., Levine, B., & Bainbridge, W. A. (2026, Nov). Behavioral change is not a unique signature of change in memory organization. *Annual Meetings of Psychonomic Society*, San Diego, CA, US.
6. **Peng, S.**, Buchsbaum, B. R., & Levine, B. (2026, Mar). Hippocampal-cortical connectivity captures traits in autobiographical memory. *Annual Meetings of Cognitive Neuroscience Society (CNS)*, Vancouver, BC, CA.
5. **Peng, S.**, & Bainbridge, W. A. (2025, May). Image memorability predicts widespread virality on social media. *Annual Meetings of Vision Science Society (VSS)*, St. Pete Beach, FL, US.
4. **Peng, S.**, & Levine, B. (2025). Cortical thickness and volume differences in individuals with severely-deficient autobiographical memory. *Annual Meetings of Cognitive Neuroscience Society (CNS)*, Boston, MA, US.
3. Feminella, J., **Peng, S.**, & St Jacques, P. (2024, Apr). Dimensionality affects memory for events in immersive virtual reality. *Annual Meetings of Cognitive Neuroscience Society (CNS)*, Toronto, ON, CA.
2. **Peng, S.**, & Dixon, P. (2022, July). Mind wandering and temporal focus on task switching. *Annual Meetings of Canadian Society for Brain, Behavior and Cognitive Science (CSBBCS)*, Halifax, NS, CA.
1. **Peng, S.**, & Dixon, P. (2022, Apr). Involvement of mind wandering in the inconsistent contradiction effect during narrative reading. *Royce Harder Conference*, Edmonton, AB, CA.

GRANTS, AWARDS, & HONORS

2026	Data Access Grant, Data Sciences Institute (\$10,000)
2026	Men's Service Group Graduate Fellowship, Baycrest (\$2,250)
2025-26	Jack & Rita Catherall Travel Fund, Baycrest (\$500)
2025	Max & Ruth Wiseman Graduate Fellowship, Baycrest (\$2,300)
2023	Maroon Research Scholarship, University of Chicago (\$30,000)
2023	M.A. Director Scholarship, University of Chicago (\$2,000)
2023	Dean's Silver Medal in Science, University of Alberta
2022	Clare Patershuk Travel Award, University of Alberta (\$750)
2021-23	Dean's Honor Roll in Science, University of Alberta

TEACHING

Course instructor

2026	Human Memory – PSY372, Department of Psychology, University of Toronto
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Teaching assistant

2026	Human Memory – PSY372, Department of Psychology, University of Toronto
2025	Statistics II – PSY202, Department of Psychology, University of Toronto
2025, 26	Introductory Psychology – PSY100, Department of Psychology, University of Toronto
2024,25,27	Statistics I – PSY201, Department of Psychology, University of Toronto

MENTORING

2026	Leo Zhang (Psychology undergraduate, summer program, University of Waterloo)
2025	Rivka Zeng-van Klei (Neuroscience undergraduate, summer program, York University)

ACADEMIC COMMUNITY SERVICES

Ad hoc review

Acta Psychologica, Journal of Experimental Psychology: General (co-review)

Other services

2026	<i>Graduate Advisor</i> , Inkblot: Undergraduate Psychology Journal, University of Toronto
2026	<i>Research Area Representative (Cognitive Neuroscience)</i> , UofT Graduate recruitment symposium
2025	<i>Planning Committee</i> , Rotman Research Institute Annual Research Conference
2024-	<i>Member</i> , Cognitive Science Research Community (CoRC), University of Toronto
2024-26	<i>Fundraising & Social Committee</i> , Annual Meetings of Toronto Area Memory Group (TAMeG)

RESEARCH SOFTWARE

Peng, S. (2025). LongiGate: An automatic gatekeeper for online longitudinal studies. *Zenodo*.
<https://doi.org/10.5281/zenodo.15253316>

TECHNICAL SKILLS

<i>Coding</i>	Python (PsychoPy, PyTorch), R, MATLAB, BASH, JavaScript (jsPsych), HTML/CSS
<i>Compute</i>	HPC/SLURM, Cloudflare (ETL workflow)
<i>Software</i>	FreeSurfer, SPM12, AFNI, FSL, Pavlovia, Prolific, Qualtrics, Vuze VR Studio, Unity
<i>Imaging</i>	Functional/structural MRI, EEG/ERP

CERTIFICATES

2023	Computational Neuroscience, Neuromatch Academy
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REFERENCES

Brian Levine, Ph.D., C. Psych., ABPP-cn, FRSC
Professor, University of Toronto | Senior Scientist, Rotman Research Institute
Ph.D. Thesis Supervisor
blevine@research.baycrest.org

Bradley Buchsbaum, Ph.D.
Professor, University of Toronto | Senior Scientist, Rotman Research Institute
Ph.D. Thesis Co-supervisor
bbuchsbaum@research.baycrest.org

Wilma Bainbridge, Ph.D.
Associate Professor, University of Chicago
Master's Thesis Main Supervisor
wilma@uchicago.edu